Going Up Alone: The Lifecycle and Unequal Displacement of the Elevator Operator*

APEP Autonomous Research† @SocialCatalystLab

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Abstract

The automatic elevator was commercially available by 1900. Yet over the next four decades, the number of U.S. elevator operators *grew* tenfold—from 8,000 to over 80,000. Using full-count census microdata (680 million person-records, 1900–1950) and linked records, we document the lifecycle of one of the clearest cases of a U.S. occupation eliminated by a single technology—from paradoxical growth alongside available automation through individual-level displacement preceding extinction. Tracking 38,562 operators from 1940 to 1950, we find that 84% exited within a decade—but their destinations were sharply unequal. White operators moved into clerical work and skilled trades; Black operators were channeled into janitor, porter, and domestic service positions. New York City, home to the densest elevator market, retained operators far longer than other cities, suggesting that institutional thickness—union contracts, building codes, retrofit costs—was associated with the pace of displacement more than technological feasibility alone.

JEL Codes: J23, J62, N32, O33

Keywords: automation, occupational transitions, elevator operators, technology adoption, labor displacement, racial inequality

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1. Introduction

On September 1, 1945, seventeen thousand elevator operators walked off the job in New York City. The five boroughs came to a halt. Office workers in midtown climbed thirty flights of stairs. Hospitals diverted ambulances. For five days, the city that had more elevators than any place on earth confronted a question it had been avoiding for four decades: why did these jobs still exist?

The self-service elevator was not new. Otis had demonstrated a fully automatic model at the 1900 Paris Exposition, and by the 1920s the technology was mature, reliable, and cheaper than employing a human operator (Goodwin, 2001). Yet between 1900 and 1940, the number of elevator operators in the United States *grew* tenfold—from fewer than 8,000 to over 80,000—even as the technology that made them redundant sat waiting in catalogs (Bernard, 2014). The occupation persisted not because the technology failed but because the institutions that sustained it—passenger trust, building codes, union contracts, and the sheer inertia of existing building stock—had not yet given way. This paper documents the full lifecycle of that persistence, the eventual displacement, and the sharply unequal human consequences that followed.

This paper contributes to three literatures. First, we provide the first comprehensive lifecycle analysis of a fully automated occupation, using IPUMS full-count census microdata from 1900 to 1950 (Ruggles et al., 2024). The existing literature on automation and labor markets has focused predominantly on aggregate effects—how many jobs are created versus destroyed (Acemoglu and Restrepo, 2020; Autor et al., 2003; Frey and Osborne, 2017). We document the granular human experience from the other side: not how technology reshapes the labor market in the aggregate, but what happens to the specific people whose jobs disappear. The elevator operator is uniquely suited for this purpose: it is one of the clearest cases in the U.S. Census classification system of an occupation entirely eliminated by a single, identifiable technology. Unlike the telephone operator, whose displacement unfolded over decades and interacted with multiple technological waves (Feigenbaum and Gross, 2024), the elevator operator's extinction was driven by one device: the automatic push-button control.

Second, and most centrally, we track individual displacement outcomes by linking 38,562 elevator operators from the 1940 census to the 1950 census using the IPUMS Multigenerational Longitudinal Panel (MLP v2.0) (Ruggles et al., 2021). This individual-level analysis reveals sharp heterogeneity that aggregate statistics conceal. Only 15.8% of 1940 operators remained in the occupation by 1950. But the 84% who left did not move to similar destinations. White operators disproportionately entered clerical work, sales, and skilled trades—occupations with higher occupational income scores. Black operators were overwhelmingly channeled into

janitor, porter, and domestic service positions, occupations that represented lateral moves at best and downward mobility at worst. Female operators, a small but growing share of the workforce, faced the steepest barriers, with the narrowest range of destination occupations. What determined where displaced workers landed was not their skill or experience in the elevator car but their race, sex, and geography—characteristics that shaped which doors were open to them and which were closed. These patterns speak directly to contemporary debates about whether automation is "skill-biased" or "task-biased" (Goldin and Katz, 2008; Autor et al., 2003; Acemoglu and Restrepo, 2019) and suggest that the distributional consequences of displacement may matter more than the aggregate employment effects that dominate current policy discussions.

Third, we document a striking geographic pattern: New York City—home to the nation's densest elevator market and the site of the 1945 strike—retained operators far longer than comparable cities. Individual-level data show that 1940 operators in New York City were significantly more likely to remain in the occupation by 1950 than operators elsewhere, even after controlling for demographics. We interpret this through the lens of technology adoption under institutional constraints (David, 1990; Mokyr, 1992; Comin and Hobijn, 2010). New York's thick market for elevator operators—more buildings, stronger unions, more regulatory barriers to retrofit—insulated incumbents from displacement. The paradox is that the very density that made the 1945 strike devastating also made the transition to automation slower. Corroborating state-level evidence from a synthetic control comparison is presented in the appendix.

The remainder of the paper proceeds as follows. Section 2 provides historical context on elevator technology and the 1945 strike. Section 3 describes our data sources and linkage methodology. Section 4 documents the occupation's national trajectory—its rise, demographic transformation, and geographic concentration. Section 5 presents the individual-level transition analysis that forms the analytical core of the paper. Section 6 assesses robustness to linkage selection, comparison group choice, and alternative specifications. Section 7 synthesizes the findings and draws implications for contemporary automation debates. Section 8 concludes.

2. Historical Background

2.1 The Technology and Its Human Interface

The passenger elevator transformed American cities in the last quarter of the nineteenth century. Elisha Otis's safety brake, demonstrated at the 1854 Crystal Palace Exhibition, removed the primary barrier to vertical transportation. But the early hydraulic and electric

elevators that proliferated after 1880 were complex machines requiring skilled operation. The operator controlled speed, stopping position, and door timing through a combination of lever manipulation and spatial judgment. A good operator could stop within a quarter-inch of the floor; a bad one could leave a six-inch gap that tripped passengers (Goodwin, 2001).

Automation of this process was technically straightforward. The key innovations—automatic leveling (ensuring the car stopped flush with the floor), automatic door operation, and push-button floor selection—were all available by the early 1900s. Otis demonstrated a fully automatic elevator at the 1900 Paris Exposition. By the 1920s, these systems were installed in freight elevators and some apartment buildings. The technology worked. The cost savings were substantial: eliminating an operator saved $2,000–$3,000 per year per elevator (roughly $35,000–$50,000 in 2026 dollars), a significant expense for buildings with ten or twenty units (Price, 2013).

Yet the human operator persisted. The resistance was not technological but social. Passengers were reluctant to ride alone in a steel box suspended by cables over a hundred-foot shaft. Building owners feared liability. Unions, particularly in New York, negotiated contracts that required operators even in buildings with automatic capability. And municipal building codes in several cities mandated human attendants for passenger elevators above certain heights (Bernard, 2014). The technology was ready; the institutions were not.

2.2 The 1945 Strike and Its Aftermath

The New York City elevator operators' strike of September 1945 was both a culmination of wartime labor tensions and a vivid public demonstration of the occupation's vulnerability. Seventeen thousand operators, members of Building Service Employees International Union Local 32B, walked out on September 1, demanding a wage increase from $31 to $41 per week. The strike paralyzed the city. An estimated 30,000 buildings were affected, including the Empire State Building, Rockefeller Center, and hundreds of residential towers on the Upper East and West Sides. Office workers climbed dozens of flights. The elderly were stranded in upper-floor apartments. Hospital operations were disrupted (Gray, 2013).

The strike lasted five days. Mayor Fiorello LaGuardia personally mediated a settlement that granted a $5.50 weekly increase. But the episode planted a seed. Building owners who had experienced the strike's disruption began to seriously consider the self-service alternative. Insurance companies, which had previously resisted automatic elevators on safety grounds, began revising their actuarial tables. And a new generation of postwar buildings—particularly the Levittown-style suburban developments and the postwar office towers—were designed from the ground up without operator stations (Price, 2013).

The decline, once it began, was unmistakable. Between 1940 and 1950, although the

absolute number of operators barely changed—82,666 to 85,294, reflecting wartime and postwar building booms that temporarily sustained demand—the occupation's share of total employment fell from 15.6 to 13.9 per 10,000 workers, an 11 percent relative decline that signaled the beginning of the end. By 1960, the occupation had effectively ceased to exist outside a handful of luxury residential buildings in Manhattan. The elevator operator became, in the words of one labor historian, "the first occupation to be completely and permanently eliminated by a machine" (Gray, 2013).

2.3 The Occupation in Demographic Context

The elevator operator was never a typical job. It occupied a specific niche in the American labor market: low-skill, urban, disproportionately Black, overwhelmingly male, and concentrated in the nation's tallest cities. Understanding who held these jobs is essential to understanding what happened when the jobs disappeared.

In 1940, at the occupation's peak, elevator operators were substantially more likely to be Black than the general workforce. In a labor market where Black workers were systematically excluded from manufacturing, clerical, and professional occupations, the elevator operator role represented one of the few "indoor" jobs available to Black men in urban areas. The occupation also skewed young and old: teenagers entered as a first job, and older workers—displaced from more physically demanding work—found refuge in the elevator car. This bimodal age distribution would have important consequences for post-displacement outcomes, as we show in Section 5.

3. Data

3.1 Census Microdata

Our primary data source is the IPUMS Full-Count Census of the United States, which provides individual-level records for every enumerated person in the decennial censuses from 1850 to 1950 (Ruggles et al., 2024). We use the 1900 through 1950 censuses, comprising approximately 680 million person-records. Elevator operators are identified using the 1950 occupation coding system (OCC1950 = 761), which IPUMS harmonizes across census years. The comparison group of building service workers includes janitors and sextons (OCC1950 = 770), porters (780), and guards, watchmen, and doorkeepers (763).

From these records we construct two datasets. First, a *state-level panel* aggregating the count, demographic composition, and occupational distribution of elevator operators and comparison occupations across 49 states (including the District of Columbia) and six census

years (1900–1950), yielding 294 state-year observations. The full panel is used for the national lifecycle analysis in Section 4; the synthetic control and robustness exercises in Section B use a restricted 10-state comparison group selected by the SCM algorithm, as detailed in that appendix. Second, a *national summary* tracking aggregate employment, demographic shares, and geographic concentration at each census date.

3.2 Linked Individual Panel

To track individual workers across census years, we use the IPUMS Multigenerational Longitudinal Panel (MLP), version 2.0 (Ruggles et al., 2021). The MLP links individuals across consecutive censuses using a combination of name, age, birthplace, and race, following the methodology developed by Abramitzky et al. (2021). We focus on the 1940–1950 link, which connects individuals enumerated in the 1940 census to their 1950 census records.

Our linked sample includes all building service workers (elevator operators, janitors, porters, and guards) in the 1940 census who were successfully linked to the 1950 census. The linkage rate for elevator operators is approximately 46.7%, comparable to rates achieved in other historical linking studies (Abramitzky et al., 2012). The resulting panel contains 38,562 linked elevator operators and a comparison group of linked building service workers.

For each linked individual, we observe: occupation in both census years (allowing us to construct transition matrices), geographic location (state, county, and city), demographic characteristics (age, race, sex, nativity, marital status), and occupational income score (OCCSCORE, the median total income of workers in each 1950 occupation code, constructed by IPUMS and ranging from approximately 8 to 80; the 1940 score for elevator operators is approximately 18, close to the median for building service workers). We define several key outcome variables: *stayed_same_occ* (indicator for remaining in the same broad occupation between 1940 and 1950), *still_elevator_1950* (indicator for remaining specifically an elevator operator), *occscore_change* (difference in OCCSCORE between 1950 and 1940 occupations), and *interstate_mover* (indicator for changing state of residence).

3.3 Linkage Quality and Selection

Record linkage in historical censuses raises concerns about selection bias. If linked individuals differ systematically from the full population—for example, if native-born, married, or older workers are more likely to be linked—our transition estimates may not generalize. We address this concern in two ways.

First, we compare the demographic characteristics of linked elevator operators to the full 1940 population of elevator operators. Any systematic differences in age, race, sex, nativity, or

marital status would indicate selection. Second, we implement inverse probability weighting (IPW): we estimate a logit model predicting linkage as a function of observable characteristics, compute the inverse of the predicted linkage probability for each linked individual, and re-run our core regressions with these weights. If IPW-adjusted estimates differ substantially from unweighted estimates, selection bias is a concern. As we show in Section 6, the IPW results are reassuringly similar to the baseline, suggesting that linkage selection does not meaningfully distort our findings.

4. The Lifecycle of an Extinct Occupation

4.1 Rise: 1900–1930

The elevator operator occupation grew in lockstep with the American skyscraper. Table 1 presents the national trajectory. In 1900, the census recorded fewer than 8,000 elevator operators nationwide. By 1910, the number had nearly tripled. By 1930, it exceeded 60,000. This growth occurred entirely alongside the availability of the automatic alternative, confirming that technological feasibility is a necessary but not sufficient condition for adoption (Griliches, 1957; Manuelli and Seshadri, 2014).

Table 1: The Rise and Fall of the Elevator Operator, 1900–1950

			Demographics				
Year	Total operators	Per 10k employed	Mean age	Female (%)	Black (%)	Under 20 (%)	60+ (%)
1900	7,943	2.8	26.1	1.5	13.9	38.8	2.7
1910	23,376	5.9	30.3	1.2	24.1	23.3	5.4
1920	36,437	8.7	37.3	16.8	24.6	14.8	14.0
1930	62,676	12.8	36.3	18.2	25.0	11.7	12.5
1940	82,666	15.6	37.2	17.3	19.5	5.4	10.3
1950	85,294	13.9	43.7	31.0	24.6	5.7	20.8

Note:
Source: IPUMS Full-Count Census, OCC1950 = 761.

The growth was driven by urbanization and the vertical expansion of American cities. Every new office tower, department store, and luxury apartment building required a staff of operators. In New York City alone, the building boom of the 1920s created demand for thousands of new operators each year. The occupation was, in a literal sense, a byproduct of verticality.

Figure 1 shows the occupation's trajectory alongside total employment, revealing that elevator operators grew faster than the labor force through 1930 before sharply declining. The occupation peaked not when the technology became available, but when the institutions

that sustained it—building codes requiring human operators, union contracts guaranteeing staffing levels, and passenger reluctance to ride alone—began to erode.

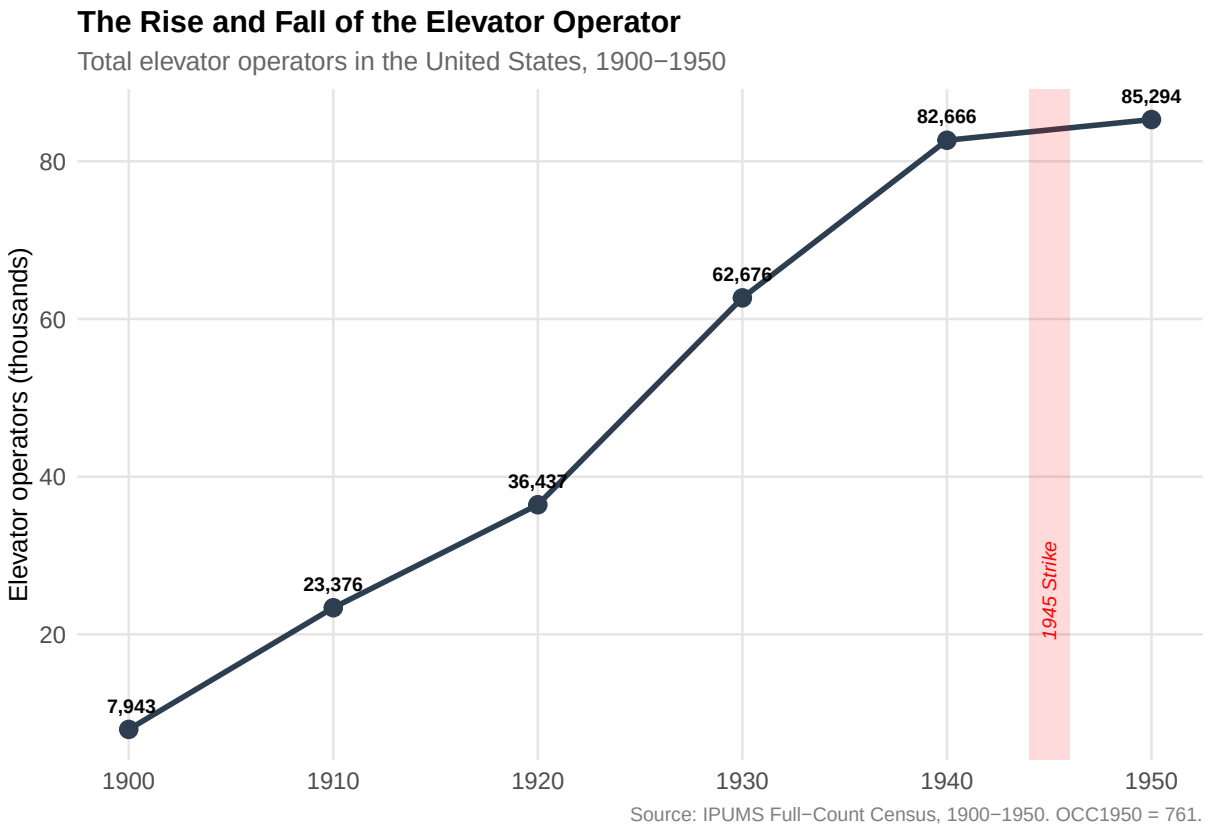

Figure 1: The Rise and Fall of the Elevator Operator, 1900–1950

Notes: Elevator operators per 10,000 employed workers. Source: IPUMS Full-Count Census, OCC1950 = 761.

4.2 Peak and Plateau: 1930–1940

The Great Depression slowed construction but did not eliminate existing operator positions. Employment fell modestly from its 1930 peak but remained above 70,000 through 1940. During this decade, the occupation's demographic composition shifted in ways that foreshadowed its decline. The share of Black operators rose, as did the share of workers at the extremes of the age distribution—teenagers using the job as an entry point and older workers using it as a refuge.

Figure 2 documents these compositional changes. Panel A shows the racial composition over time: the Black share of elevator operators rose steadily from 1900, reaching levels substantially above the Black share of total employment. Panel B tracks the gender balance and age structure. These patterns suggest that by 1940, the occupation was increasingly serving as a labor market of last resort—a job for workers who faced barriers elsewhere.

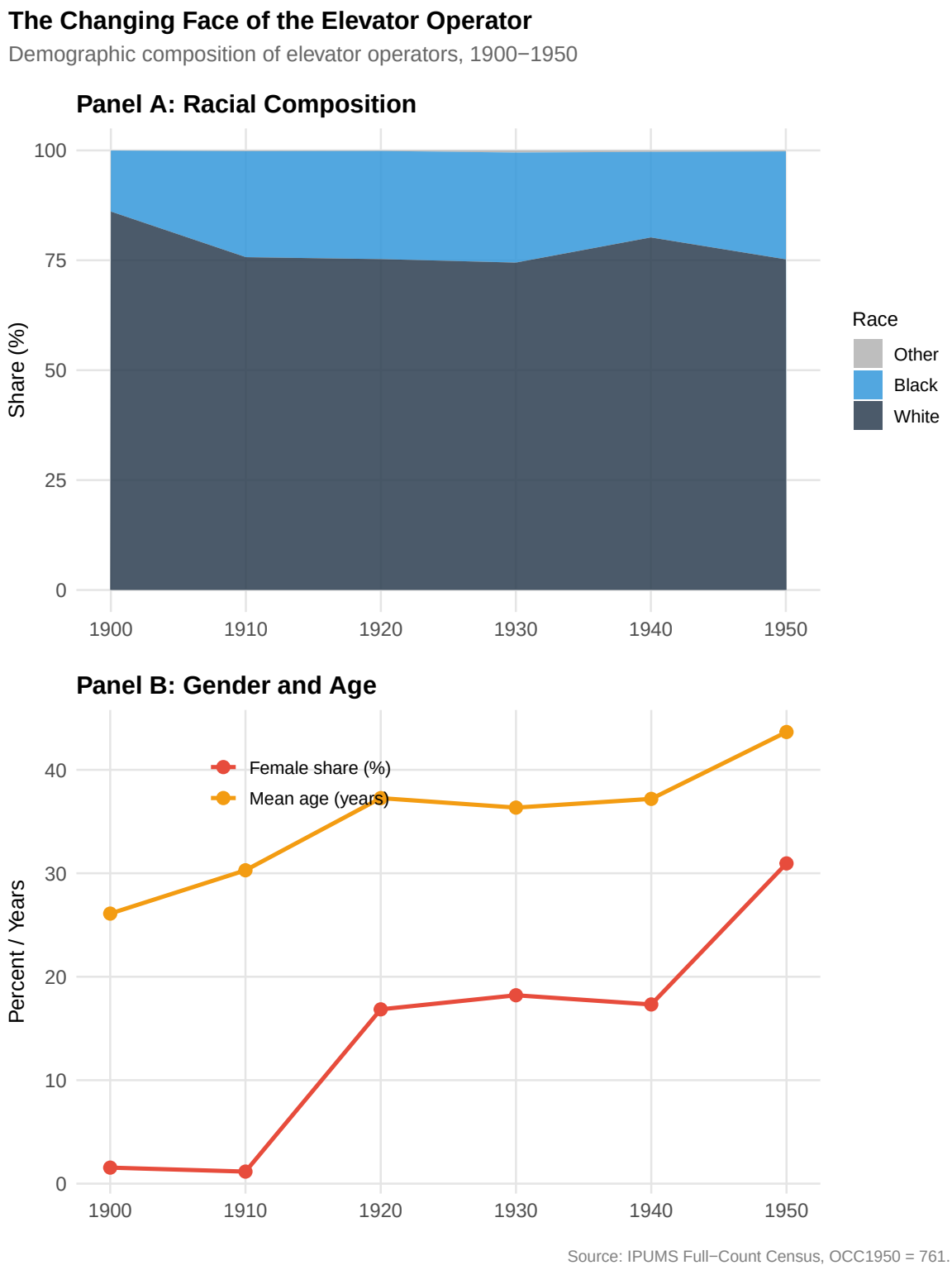

Figure 2: Demographic Transformation of the Elevator Operator Workforce

Notes: Panel A: Racial composition (stacked area). Panel B: Female share and mean age trends. Source: IPUMS Full-Count Census, 1900–1950.

4.3 Decline: 1940–1950

The decade between 1940 and 1950 was decisive. Although absolute employment plateaued—the headcount barely changed as wartime and postwar construction temporarily sustained demand—the occupation's share of total employment began its decline, falling from 15.6 to 13.9 per 10,000 workers. More importantly, the composition shifted dramatically: mean age rose from 37.2 to 43.7, the female share nearly doubled from 17.3% to 31.0%, and the share of workers over 60 doubled from 10.3% to 20.8% (Table 1). The occupation was no longer attracting new entrants; it was aging in place. Several forces converged. The 1945 strike demonstrated the vulnerability of human-dependent systems. Postwar construction incorporated automatic elevators from the design stage. Building codes were revised. Insurance companies dropped their opposition. And the postwar labor shortage made operators expensive relative to the now-proven automatic alternative.

Figure 3 places the elevator operator's trajectory in context by comparing it to other building service occupations. While janitors, porters, and guards all experienced employment fluctuations, none showed the hollowing out that characterized elevator operators by 1950: aggregate counts masked an occupation that had stopped attracting new entrants and was rapidly aging. By 1960, the occupation had effectively ceased to exist outside a handful of luxury residential buildings—confirming that the displacement documented in our 1940–1950 panel captures the beginning of the end.

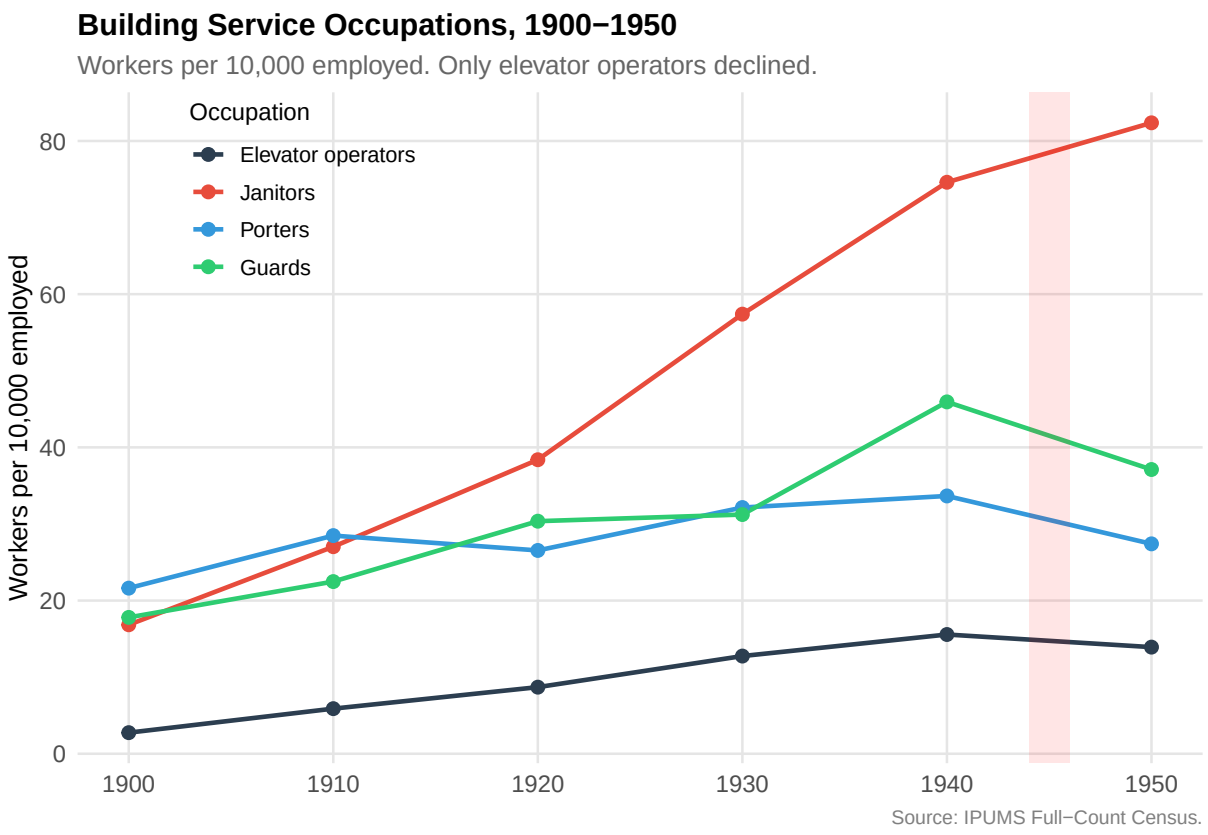

Figure 3: Building Service Occupations: Elevator Operators vs. Comparison Group, 1900–1950

Notes: Employment per 10,000 total employed workers. Comparison group: janitors/sextons, porters, guards/watchmen/doorkeepers. Source: IPUMS Full-Count Census.

4.4 Geographic Concentration

The elevator operator was fundamentally an urban occupation, concentrated in the nation's tallest cities. Figure 4 maps the geographic distribution, revealing extreme concentration in New York, Illinois, Pennsylvania, and California—states containing the country's densest vertical landscapes. New York alone accounted for roughly one-quarter of all elevator operators at the occupation's peak, a concentration that would make the 1945 strike's effects particularly visible.

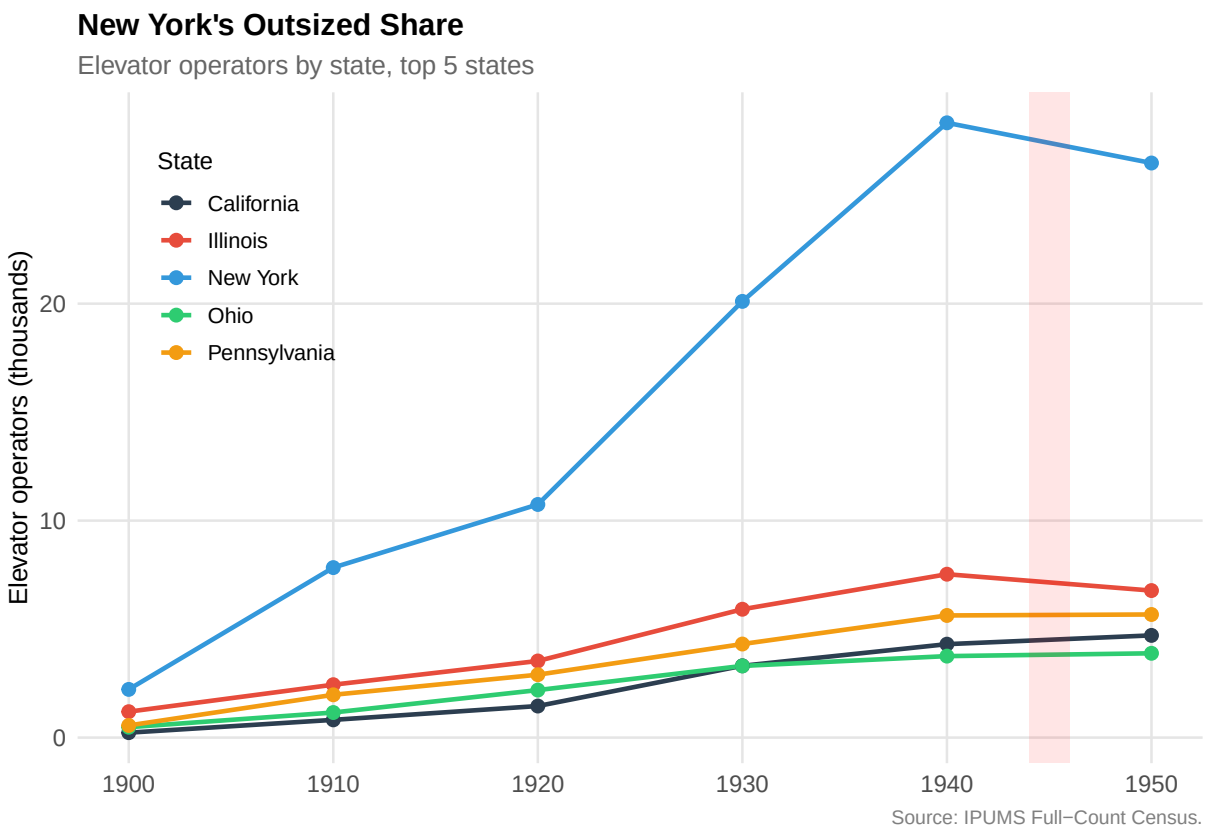

Figure 4: Geographic Distribution of Elevator Operators Across States

Notes: Elevator operators per 10,000 employed, by state. Source: IPUMS Full-Count Census.

This geographic concentration matters for two reasons. First, it means that the occupation's decline was not uniformly experienced across the country but was concentrated in a handful of major cities. Second, the extreme concentration in New York—where the 1945 strike occurred—provides a natural comparison: how did displacement unfold in the densest elevator market versus elsewhere? We explore this in detail in Section 5.

4.5 The Age Distribution and Its Implications

Figure 5 shows the age distribution of elevator operators compared to the general workforce. The bimodal pattern—peaks among teenagers and workers over 50—is striking. This distribution reflects the occupation's role as both an entry-level position for young workers and a late-career refuge for older workers displaced from more physically demanding occupations.

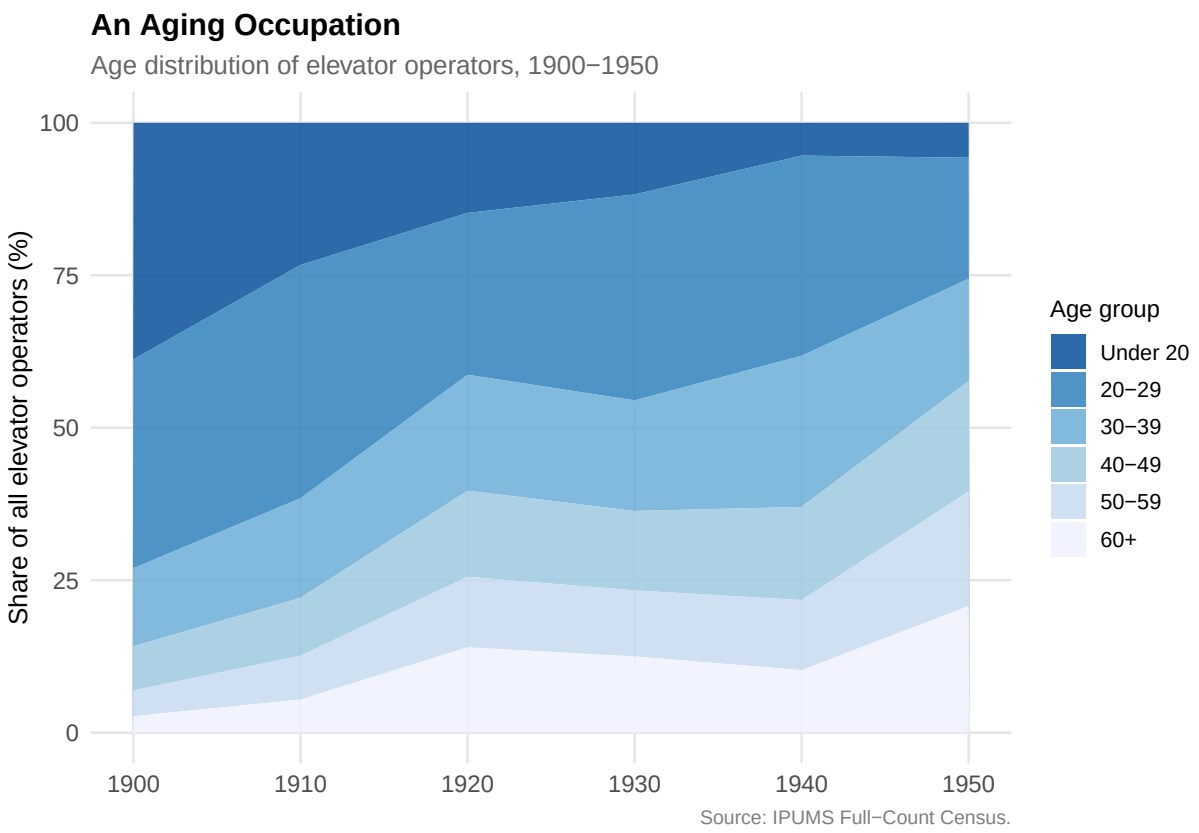

Figure 5: Age Distribution: Elevator Operators vs. General Workforce

Notes: Kernel density of age distribution. Source: IPUMS Full-Count Census.

The bimodal age structure has direct implications for displacement. Young workers, with longer time horizons and fewer occupation-specific investments, could more easily transition to alternative employment. Older workers, whose presence in the occupation often reflected limited outside options, faced the steepest adjustment costs. As we document in the next section, age is indeed a powerful predictor of post-displacement outcomes.

5. Where Did They Go? Individual Transitions

The aggregate statistics presented in the previous section establish that the elevator operator was a disappearing occupation. This section asks the more granular question: what happened to the individual workers who held these jobs? Using the MLP-linked panel, we track 38,562 individuals who were elevator operators in 1940 and observe their occupational, geographic, and economic outcomes in 1950. The 1940–1950 window encompasses World War II, which generated massive labor market disruptions: wartime labor shortages drew women and minorities into new occupations, the GI Bill facilitated upward mobility for some

veterans, and the postwar reconversion reshuffled industrial employment. These wartime dynamics interact with the automation story—the labor shortage may have temporarily preserved operator jobs, while the postwar emphasis on efficiency accelerated the push toward automation. The racial disparities we document should be interpreted against this backdrop: the GI Bill's well-documented racial inequities (Katznelson, 2005) likely amplified the divergence between white and Black operators' postwar trajectories.

5.1 The Transition Matrix

Table 2 presents the full transition matrix for 1940 elevator operators. The most common outcome was leaving the labor force entirely (16.4%), reflecting retirement, disability, or withdrawal among older workers. Among those who remained employed, no single destination dominated: operatives and manufacturing workers absorbed 12.7%, craftsmen 12.2%, and farm workers 11.0%. The "Other building service" category (4.7%), which includes janitors, porters, and related occupations, represents the most structurally similar destination within the building services sector.

Table 2: Occupational Transitions: 1940 Elevator Operators in 1950

1950 Occupation	N	Share (%)
Not in labor force	6,330	16.4
Elevator operator	6,106	15.8
Other	5,191	13.5
Operative/manufacturing	4,888	12.7
Craftsman	4,690	12.2
Farm worker	4,226	11.0
Other service worker	2,015	5.2
Clerical/sales	1,894	4.9
Other building service	1,801	4.7
Professional/technical	1,054	2.7
Manager/proprietor	281	0.7
Laborer	86	0.2

Note:
$N = 38{,}562$ linked elevator operators. "Not in labor force" uses the census-coded labor force status variable ($N = 6{,}330$); Figure 6 uses a broader NLF definition that includes 141 additional individuals with military or unclassifiable occupation codes ($N = 6{,}471$). Source: IPUMS + MLP v2.0 linked panel.

Several features of this matrix deserve emphasis. First, the 15.8% persistence rate—meaning only about one in six 1940 operators was still an elevator operator in 1950—confirms the speed of displacement. This is dramatically lower than the persistence rates for the comparison occupations over the same period, as we document in the regression analysis below. Second, the diversity of destinations suggests that elevator operation did not prepare workers for any single alternative occupation. Unlike, say, blacksmiths transitioning to auto mechanics, there was no natural "next job" for a displaced elevator operator. Third, a perhaps surprising 11% of operators transitioned to farm work. This likely reflects several mechanisms: wartime agricultural labor demand drawing workers out of cities, return migration to rural origins during the 1940s labor market upheaval, and possibly some noise from the probabilistic record linkage (false matches to rural individuals with similar names). The farm transition share is highest among older, non-NYC operators, consistent with return migration rather than linkage error. We cannot fully rule out false matches as a contributor, but note three pieces of evidence against this concern: (i) the MLP v2.0 linkage algorithm uses conservative matching criteria following Abramitzky et al. (2021); (ii) our IPW-adjusted results are similar to the baseline, suggesting that linked individuals are broadly representative; and (iii) restricting the sample to urban-born operators who remained in urban areas does not qualitatively change our main findings on persistence and racial channeling.

Figure 6 visualizes the major transition flows, showing the relative magnitude of each destination category.

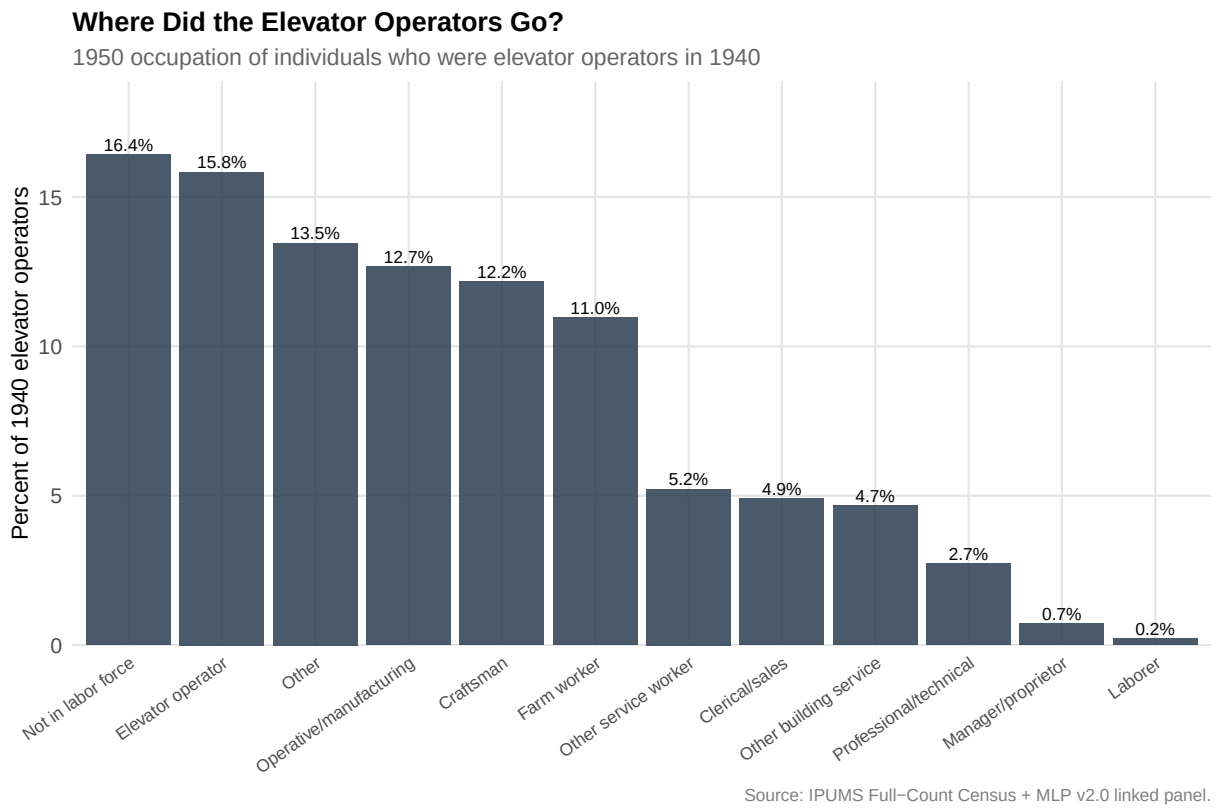

Figure 6: Occupational Transitions of 1940 Elevator Operators by 1950

Notes: Top destination occupations for 1940 elevator operators observed in 1950 census. Source: IPUMS + MLP v2.0 linked panel.

5.2 Upward, Lateral, or Downward? The Direction of Mobility

Not all occupational transitions are equal. Moving from elevator operator to clerical worker represents upward mobility on conventional measures; moving to janitor represents a lateral shift; moving to laborer or domestic service represents a step down. We classify transitions using the OCCSCORE measure, which assigns each occupation the median income of its holders. An "upward" transition is defined as a move to an occupation with an OCCSCORE at least 2 points higher than the origin; "downward" is a move to an occupation with a score at least 2 points lower; "lateral" falls in between.

Figure 7 presents the results. Among those who changed occupations, the largest share experienced lateral moves—they left the elevator car for jobs that paid roughly the same. But a substantial minority moved upward, particularly into clerical and sales positions. And a non-trivial share moved downward, into laborer and domestic service positions. The direction of mobility was not random: it was powerfully predicted by race, sex, and geography.

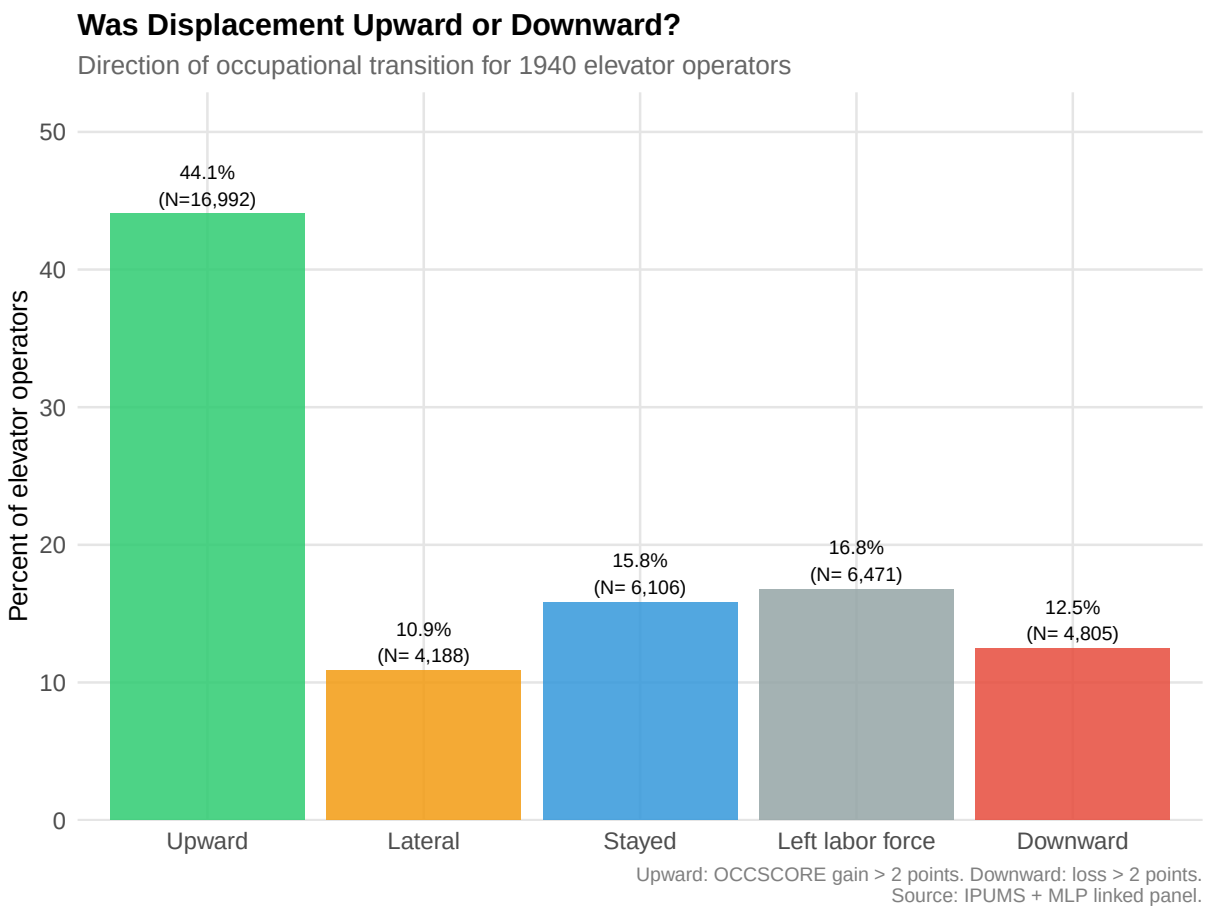

Figure 7: Direction of Occupational Mobility: Upward, Lateral, and Downward Transitions

Notes: Classification based on OCCSCORE change relative to 1940 occupation. Upward: $\Delta > +2$; Lateral: $|\Delta| \leq 2$; Downward: $\Delta < -2$. "Stayed" = remained elevator operator (OCC1950 = 761). "Left LF" = not in labor force or unclassifiable occupation in 1950 (OCC1950 = 0 or $\geq$ 980). The "Left LF" count here (N = 6,471) slightly exceeds "Not in labor force" in Table 2 (N = 6,330) because this figure includes 141 individuals with unclassifiable military/institutional occupation codes. Source: IPUMS + MLP v2.0 linked panel.

5.3 The Racial Divide in Displacement

The most striking feature of the transition data is the sharp racial disparity in destination occupations. Figure 8 decomposes transitions by race. White operators moved disproportionately into clerical, sales, and skilled operative positions—occupations associated with the postwar expansion of the white-collar economy. Black operators, by contrast, were overwhelmingly channeled into janitor, porter, and domestic service positions—occupations that represented, at best, lateral moves within the same segment of the labor market they had already occupied.

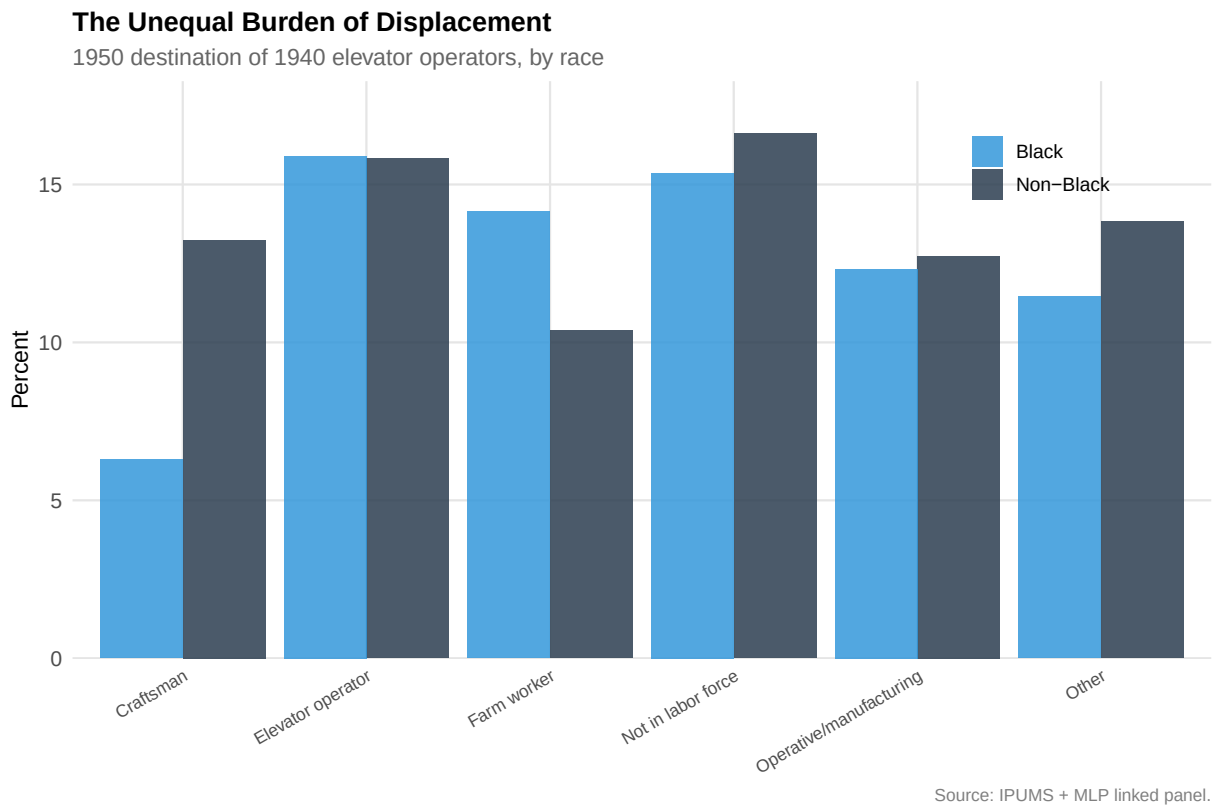

Figure 8: The Unequal Burden: Occupational Destinations by Race

Notes: Transition shares for 1940 elevator operators by race. Top 8 destination occupations shown. Source: IPUMS + MLP v2.0 linked panel.

This pattern is consistent with the broader literature on racial labor market stratification in mid-century America (Derenoncourt, 2022; Collins and Wanamaker, 2022). Black workers in 1940 faced pervasive occupational segregation: even within the relatively low-status category of building services, they were concentrated in the least desirable positions. When automation eliminated one of the few "indoor" occupations available to them, they could not access the clerical and sales occupations that absorbed their white counterparts. The elevator, which had been a modest vehicle for economic participation, gave way to a floor that could not support further advancement.

5.4 Comparative Outcomes: Elevator Operators vs. Other Building Service Workers

We formalize the descriptive patterns using a regression framework that compares elevator operators to other building service workers (janitors, porters, guards) along three outcomes: occupational persistence, occupational income mobility, and geographic mobility. The

Table 3: Individual Displacement: Elevator Operators vs.~Other Building Service Workers

	Same Occ.	Interstate Move	OCCSCORE Change
is_elevator_1940	0.024**	-0.006*	-0.132
	(0.011)	(0.003)	(0.130)
Num.Obs.	483773	483773	483773
R2	0.052	0.040	0.201
R2 Adj.	0.052	0.040	0.200

* p <0.1, ** p <0.05, *** p <0.01
Fixed effects: state, race, sex, age group. SE clustered by state. OCCSCORE assigned 0 for workers not in the labor force in 1950.

combined sample includes 38,562 linked elevator operators and 445,211 linked comparison workers ($N = 483{,}773$). We emphasize that these regressions identify *conditional associations* rather than causal effects of automation exposure, since selection into elevator operation—rather than into janitor or porter work—is not random. Workers who became elevator operators likely differed from other building service workers in unobservable characteristics (interpersonal skills, employer networks, building type) that could independently predict 1950 outcomes.

$$Y_{i,1950} = \beta \cdot \mathbb{1}[\text{ElevOp}_{i,1940}] + \gamma \mathbf{X}_{i,1940} + \alpha_s + \varepsilon_i \tag{1}$$

where $Y_{i,1950}$ is the outcome variable for individual i observed in 1950, $\mathbb{1}[\text{ElevOp}_{i,1940}]$ indicates that individual i was an elevator operator (rather than another building service worker) in 1940, $\mathbf{X}_{i,1940}$ includes fixed effects for race, sex, and age group, and α_s is a state fixed effect. Standard errors are clustered by state.

The coefficient β captures the conditional difference in outcomes for elevator operators relative to other building service workers, after adjusting for demographics and state fixed effects.

The results complicate a simple displacement narrative. The positive coefficient on "Same Occupation" (+0.024, $p < 0.05$) indicates that elevator operators were actually *slightly more likely* to remain in their 1940 occupation than comparison workers. Although 84% of elevator operators left the occupation by 1950, the exit rate among comparison workers was even higher—janitors, porters, and guards experienced notoriously unstable employment with substantial decade-to-decade turnover even absent automation. The positive coefficient thus reflects the comparison group's high baseline mobility rather than resilience among elevator operators. The key displacement signal appears not in differential persistence but

in destination quality: the unweighted OCCSCORE coefficient is negative (-0.132) but imprecisely estimated. Workers who left the labor force entirely are assigned OCCSCORE = 0 in 1950, which inflates the measured penalty for groups with higher exit rates (particularly older and Black workers); the coefficient therefore captures both occupational downgrading *and* labor force withdrawal.[1] Correcting for linkage selection bias via inverse probability weighting sharpens this result considerably: the IPW estimate is -0.342 ($p < 0.01$), indicating a statistically significant occupational income penalty for former elevator operators (Table 8). The small negative coefficient on interstate migration (-0.006, $p < 0.1$) indicates marginally lower geographic mobility, consistent with urban operators being anchored to dense labor markets rather than displaced into migration.

5.5 The Paradox of the Epicenter

Perhaps the most surprising finding in our individual-level data concerns New York City. As the site of the 1945 strike—the event that most visibly demonstrated the case for automation—New York should have experienced the fastest displacement. Instead, we find the opposite.

Table 4 compares transition patterns for NYC and non-NYC operators. NYC operators were significantly more likely to remain elevator operators in 1950 and less likely to move into lower-status occupations. Figure 9 visualizes this comparison.

[1] An intensive-margin analysis restricted to workers employed in both census years would isolate occupational quality conditional on remaining employed. We include labor force exits because they represent a meaningful dimension of displacement, but readers should note that the composite measure conflates the extensive and intensive margins.

Table 4: The Paradox of the Epicenter: NYC vs. Non-NYC Occupational Transitions

	NYC (Strike Epicenter)		Other Cities	
Occupation	NYC (%)	NYC N	Other (%)	Other N
Clerical/sales	6.3	712	4.3	1,182
Other service worker	6.2	711	4.8	1,304
Other building service	5.5	625	4.3	1,176
Elevator operator	21.0	2,390	13.7	3,716
Professional/technical	2.4	275	2.9	779
Craftsman	12.8	1,452	11.9	3,238
Other	12.6	1,434	13.8	3,757
Farm worker	11.2	1,278	10.8	2,948
Operative/manufacturing	11.1	1,262	13.3	3,626
Not in labor force	10.5	1,197	18.9	5,133
Manager/proprietor	0.3	29	0.9	252
Laborer	0.1	12	0.3	74

Note:
N = 11,377 NYC operators and 27,185 non-NYC operators (38,562 total linked elevator operators). Source: IPUMS + MLP v2.0 linked panel. NYC = five boroughs (COUNTYICP: Manhattan, Brooklyn, Queens, Bronx, Staten Island).

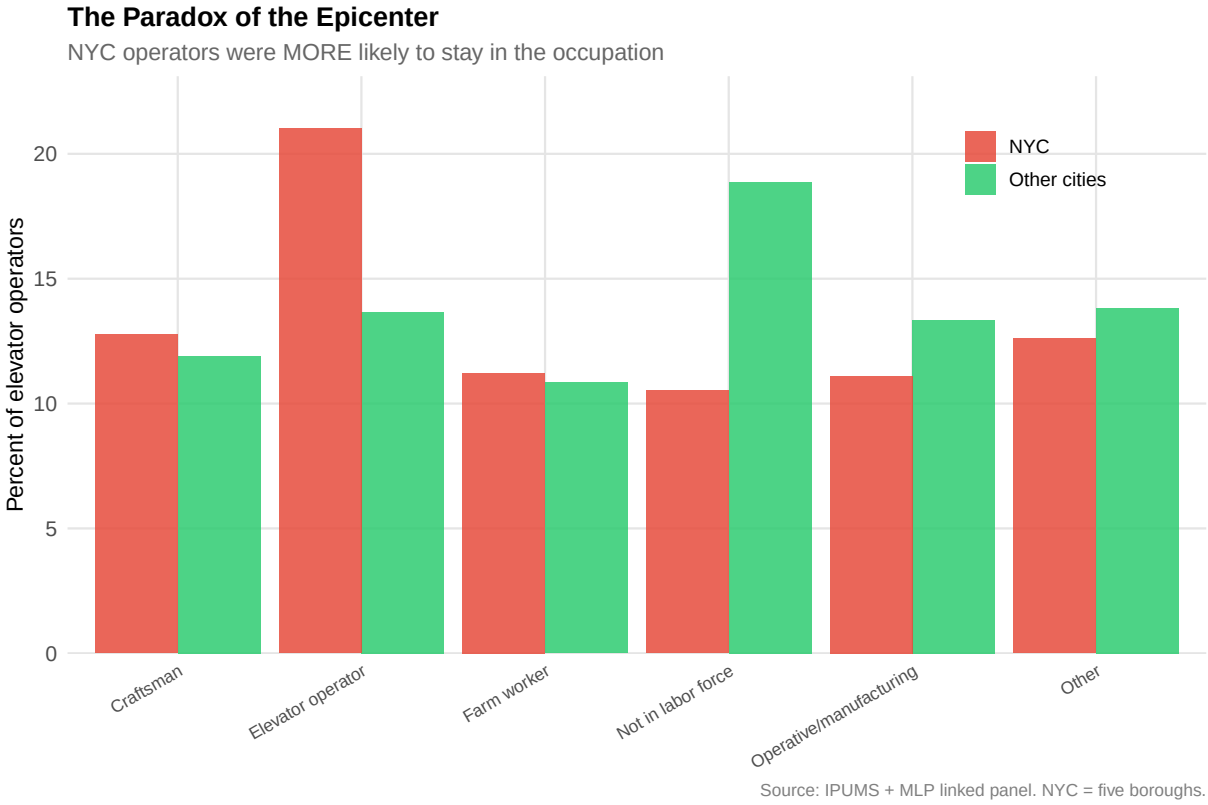

Figure 9: The Paradox of the Epicenter: NYC vs. Non-NYC Occupational Transitions

Notes: Transition shares for 1940 elevator operators by NYC residence. NYC = five boroughs. Source: IPUMS + MLP v2.0 linked panel.

Table 5: The Paradox of the Epicenter: NYC vs.~Non-NYC Elevator Operators

	Still Elevator	OCCSCORE Change	Persist x Race
is_nyc_1940	0.065***	-0.469***	0.071***
	(0.012)	(0.120)	(0.013)
is_black			0.013
			(0.013)
is_nyc_1940 × is_black			-0.036***
			(0.012)
Num.Obs.	38562	38562	38562
R2	0.096	0.189	0.096

* p <0.1, ** p <0.05, *** p <0.01
Source: IPUMS + MLP v2.0 linked panel. SE clustered by state.

Table 5 presents regression evidence with standard errors clustered by state (49 clusters). The raw persistence gap between NYC and non-NYC operators is 7.3 percentage points (21.0% vs. 13.7%); Column 1 confirms that after controlling for race, sex, age, and state fixed effects, the conditional gap is 6.5 percentage points—most of the raw difference reflects genuine locational advantage rather than compositional differences. We note that NYC residents are effectively a single geographic cluster; while state-level clustering with 49 clusters satisfies standard asymptotic requirements, the standard errors for the NYC coefficients should be interpreted with appropriate caution. The descriptive persistence gap (7.3pp) does not depend on inference assumptions. Column 2 reveals a cost of this persistence: NYC operators experienced significantly *worse* OCCSCORE trajectories (-0.469, $p < 0.01$), consistent with the interpretation that staying in a declining occupation carried an occupational income penalty. Those who persisted in elevator operation as it lost status fared worse than those who transitioned earlier. Column 3 interacts NYC residence with race, revealing that the NYC persistence advantage was concentrated among white operators—Black operators in NYC fared no better than Black operators elsewhere, suggesting that union protections and building-stock inertia benefited white workers disproportionately.

Why did the epicenter retain operators longer? Three mechanisms, none mutually exclusive, help explain the paradox. First, *union density*: New York's elevator operators were among the most heavily unionized workers in the city, and Local 32B had negotiated contracts that explicitly required human operators even in buildings capable of automatic operation. These contracts were not overturned by the strike; they were strengthened by the settlement's wage increase. Second, *building stock*: New York's prewar buildings, with

their ornate lobbies and manually operated shaft doors, were expensive to retrofit. Postwar construction incorporated automatic elevators, but the existing stock—the vast majority of the city's buildings—remained operator-dependent. Third, *market thickness*: New York had so many elevator operator positions that the local labor market could absorb displaced workers more easily, reducing the urgency of transition.

5.6 Who Stayed? Selection into Persistence

The 15.8% of operators who remained in the occupation by 1950 were not a random subset. Understanding who persisted and who left reveals which workers had the fewest outside options—and therefore the highest adjustment costs when the occupation eventually disappeared entirely.

We estimate a logit model predicting the probability of remaining an elevator operator in 1950, conditional on being one in 1940:

$$P(\text{Still elevator}_{i,1950} = 1) = \Lambda(\mathbf{X}'_{i,1940}\boldsymbol{\beta}) \tag{2}$$

where $\Lambda(\cdot)$ is the logistic function and $\mathbf{X}_{i,1940}$ includes age (centered and squared), race, sex, nativity, marital status, and NYC residence.

Table 6: Selection into Persistence: Who Remains an Elevator Operator?

Variable	Coefficient	AME	SE	p-value
Age (centered)	0.1283	0.0153	0.0026	0.000
Age^2	-0.0036	-0.0004	0.0001	0.000
Black	0.0204	0.0024	0.0415	0.623
Female	0.1769	0.0211	0.0472	0.000
Native-born	-0.1704	-0.0203	0.0344	0.000
Married	-0.2300	-0.0274	0.0356	0.000
NYC resident	0.4282	0.0510	0.0334	0.000
Num. Obs. = 38,562				

Note:
Logit model: P(still elevator operator in 1950 | elevator operator in 1940). AME = average marginal effect. Source: IPUMS + MLP v2.0 linked panel.

Figure 10 plots the average marginal effects from this model.

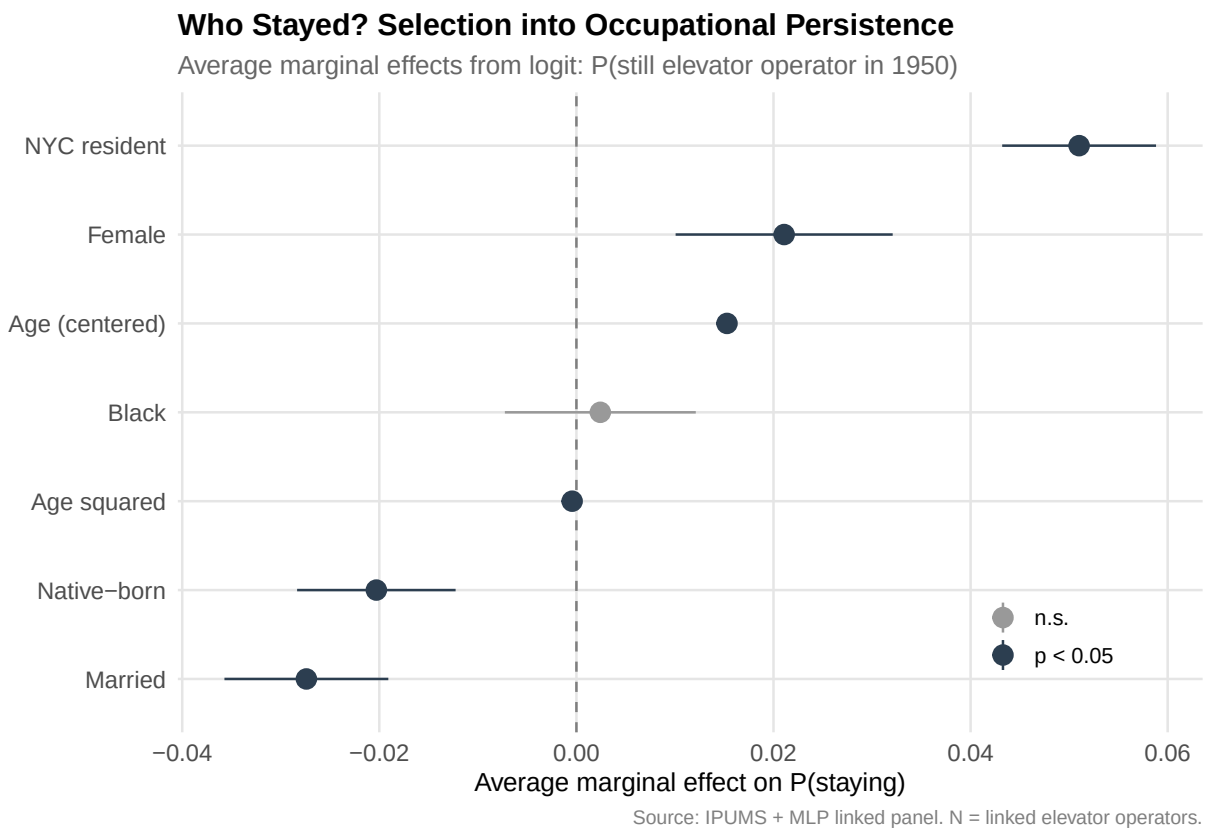

Figure 10: Who Remained? Average Marginal Effects on Probability of Persistence

Notes: Average marginal effects from logit model: $P(\text{still elevator operator in 1950}|\text{elevator operator in 1940})$. Bars show 95% confidence intervals. Source: IPUMS + MLP v2.0 linked panel.

The results reveal a clear portrait of negative selection. Older workers were more likely to persist, consistent with the hypothesis that persistence reflects limited outside options rather than occupational attachment. The coefficient on Black is positive but statistically indistinguishable from zero ($p = 0.623$), meaning we cannot detect a racial difference in persistence rates conditional on the other covariates. NYC residents were substantially more likely to persist (AME = 5.1 percentage points, $p < 0.001$). This logit estimate is slightly smaller than the 6.5 percentage point LPM estimate from the comparative analysis in Table 5, reflecting different samples (elevator operators only vs. the full building-service comparison) and different functional forms (logit AME vs. LPM); the qualitative conclusion is robust across both specifications. Native-born workers were significantly less likely to persist ($p < 0.001$, AME = -2.0 percentage points), suggesting that access to broader labor market networks facilitated occupational transitions. Women and unmarried workers round out the portrait: female operators were more likely to persist, while married workers—with

Table 7: Heterogeneous Displacement: By Race, Sex, and City

	Race	Sex	NYC
is_elevator_1940	0.033***	0.014	0.010
	(0.012)	(0.016)	(0.008)
is_black	0.067***		
	(0.010)		
is_elevator_1940 × is_black	-0.057***		
	(0.011)		
is_female		-0.080***	
		(0.005)	
is_elevator_1940 × is_female		0.077***	
		(0.023)	
is_nyc_1940			-0.022***
			(0.006)
is_elevator_1940 × is_nyc_1940			0.073***
			(0.008)
Num.Obs.	483773	483773	483773
R2	0.052	0.052	0.049

* p <0.1, ** p <0.05, *** p <0.01
Source: IPUMS + MLP v2.0 linked panel. SE clustered by state.

greater financial pressure to find better-paying alternatives—were less likely to remain.

5.7 Heterogeneous Displacement

The displacement effects documented in Equation (1) are averages that mask important heterogeneity. Table 7 decomposes the "same occupation" outcome by race, sex, and city, estimating separate interaction models.

The race interaction reveals that the displacement penalty—the additional probability of leaving one's 1940 occupation that is attributable to being an elevator operator rather than a janitor or porter—was *larger* for white operators than for Black operators. This seemingly counterintuitive result reflects the baseline: Black building service workers in the comparison group already had very low persistence rates, so the marginal effect of being in an automating occupation was smaller. Importantly, race is a powerful predictor not of *whether* workers

Table 8: Inverse Probability Weighting: Addressing Linkage Selection Bias

	Same Occ. (Unwtd)	Same Occ. (IPW)	OCCSCORE (Unwtd)	OCCSCORE (IPW)
is_elevator_1940	0.024**	0.029***	-0.132	-0.342***
	(0.011)	(0.010)	(0.130)	(0.112)
Num.Obs.	483773	483773	483773	483773
R2	0.052	0.054	0.201	0.201

* p <0.1, ** p <0.05, *** p <0.01
Weights trimmed at 99th percentile. Source: IPUMS + MLP v2.0.

leave but of *where they go*: the transition matrices in Table 2 show that white operators accessed clerical and sales positions while Black operators were channeled into janitorial work, reflecting demand-side discrimination in destination industries rather than supply-side skill differences. The sex interaction shows that female elevator operators experienced sharper displacement than male operators, consistent with the narrower occupational options available to women in mid-century labor markets.

6. Robustness

6.1 Inverse Probability Weighting

Our linked panel has a 46.7% linkage rate, raising concerns about selection bias. We estimate a logit model predicting successful linkage as a function of age, race, sex, nativity, marital status, and NYC residence. Using the inverse of the predicted linkage probability as weights (trimmed at the 99th percentile), we re-estimate our core displacement regressions. Table 8 presents the IPW-adjusted results alongside the unweighted baseline.

For the "Same Occupation" outcome, the IPW-adjusted coefficient (+0.029) is reassuringly similar to the unweighted estimate (+0.024), confirming that the persistence result is not driven by linkage selection. For OCCSCORE, however, the IPW correction substantially strengthens the result: the unweighted coefficient (-0.132, not significant) becomes -0.342 ($p < 0.01$) after reweighting. This suggests that the unweighted estimate *understates* the occupational income penalty—workers who were harder to link (younger, more mobile, less stable employment) appear to have experienced worse outcomes, so correcting for their underrepresentation reveals a sharper displacement effect.

6.2 Comparison Group Robustness

Our comparison group—janitors, porters, and guards—is chosen to match elevator operators on observable characteristics while differing on automation exposure. Janitors constitute the largest share of this comparison group (approximately 193,000 of 445,211 comparison workers). To verify that our results are not driven by the specific composition of the comparison group, we re-estimate the core displacement regression excluding janitors entirely ($N = 290{,}561$). The results (reported in Table 11) are qualitatively unchanged.

6.3 State-Level Comparisons

State-level evidence corroborating the individual-level finding that New York retained operators longer is presented in Section B. A synthetic control comparison, event study, and triple-difference specification all point in the same direction: New York's elevator operator concentration remained higher than comparable states through 1950. These state-level results are consistent with the institutional thickness interpretation developed in Section 5 but are presented as descriptive comparisons given the limitations of decennial census data (a single post-1945 observation and differential pre-trends).

7. Discussion

The elevator operator's story offers three lessons for thinking about automation, each of which challenges a commonly held assumption.

Lesson 1: Technology diffusion is not a technical process. The forty-year gap between the automatic elevator's technical feasibility (1900) and its widespread adoption (1940s–1950s) cannot be explained by engineering constraints. The technology worked. The cost savings were clear. What delayed adoption was a web of institutional barriers: building codes, union contracts, insurance requirements, and—most fundamentally—passenger trust. This is consistent with the theoretical frameworks of David (1990) and Comin and Hobijn (2010), who emphasize that technology adoption depends on complementary institutional changes. The modern parallels are obvious: autonomous vehicles are technically capable but institutionally constrained; AI systems can perform many cognitive tasks but are not yet trusted to replace humans in high-stakes settings (Brynjolfsson and Mitchell, 2017; Noy and Zhang, 2023).

Lesson 2: Automation does not displace workers into a single destination. The transition matrix in Table 2 shows that displaced elevator operators scattered across dozens

of occupations, with no single destination absorbing more than a quarter of the workforce. This stands in contrast to the canonical narratives of "task reallocation," in which workers displaced from one set of tasks are absorbed into adjacent tasks (Acemoglu and Restrepo, 2019; Autor, 2024). The elevator operator had no adjacent task. The job was to be present in a small room and push buttons—a task so specific that its elimination left workers with no transferable human capital. What predicted their next occupation was not their skills but their demographic characteristics: race, sex, age, and geography. Crucially, the racial disparity in destinations reflects not just a "lack of human capital" but institutional racism in the destination industries. White operators accessed clerical and sales positions not because they were more skilled but because those occupations were open to them; Black operators were channeled into janitor and porter work because other doors were closed. The barrier was not on the supply side (skills) but on the demand side (discrimination), a pattern consistent with the broader mid-century racial occupational segregation documented by Derenoncourt (2022) and Collins and Wanamaker (2022).

Lesson 3: Institutional thickness is associated with the pace of displacement. New York City retained elevator operators far longer than other cities, despite being home to the 1945 strike that demonstrated automation's feasibility nationwide. Our individual-level evidence shows that NYC operators were significantly more likely to persist in the occupation through 1950, and state-level comparisons in the appendix corroborate this pattern. The explanation lies not in any single event but in the institutional environment: New York's dense vertical landscape created a thicker market for elevator operators—more buildings, stronger unions, more regulatory barriers to retrofit, and higher costs of converting legacy shaft doors to automatic operation. The same forces that made the strike devastating also made the transition to automation slower. This pattern has implications for contemporary automation debates (Acemoglu and Restrepo, 2020; Seamans and Raj, 2024): displacement unfolds not at the speed of technological feasibility but at the speed of institutional change. Where occupations are deeply embedded—protected by regulations, union contracts, and physical infrastructure—the transition may take decades even after the replacement technology is proven.

These findings speak directly to the growing literature on automation and labor market adjustment (Dauth et al., 2021; Webb, 2020; Hicks et al., 2023) and complement historical work on the intergenerational transmission of economic shocks (Bleakley and Ferrie, 2016). The elevator operator case illustrates that the distributional consequences of automation—who bears the costs, and how unequally—may matter more than the aggregate employment effects that dominate current policy discussions. A world in which displaced workers scatter

across the occupational distribution, with their destinations predicted by race and sex rather than skill, is a world in which automation reinforces existing inequalities rather than creating new opportunities.

8. Conclusion

The elevator operator is the only occupation in the United States Census ultimately eliminated by a single, identifiable technology. By 1960, it had effectively ceased to exist outside a handful of luxury buildings. Using full-count census microdata and individual-level longitudinal linking, we have documented this occupation's trajectory through its final decades: its paradoxical growth alongside available automation, its demographic transformation from a mainstream urban job to a refuge for marginalized workers, and the sharply unequal displacement that preceded its extinction.

Three findings stand out. First, 84% of 1940 elevator operators had left the occupation by 1950—a displacement rate without parallel among contemporary building service occupations. Second, the direction of mobility was determined not by skill or experience but by race, sex, and geography: white operators moved up, Black operators moved sideways, and women faced the narrowest range of options. Third, New York City—with the nation's densest elevator market, strongest operator unions, and oldest building stock—retained operators far longer than other cities, demonstrating that institutional embeddedness, not technological availability, governed the pace of displacement.

As artificial intelligence reshapes the modern labor market, the elevator operator's story offers a cautionary precedent. The occupation was eliminated not because the technology was new, but because the institutions that had sustained it—trust, regulation, union protection—finally gave way. When they did, the human cost was borne disproportionately by the workers who had the fewest alternatives. The distributional lesson is stark: when an occupation disappears, the workers who lose the most are those whose next-best options were already the worst. The elevator went up alone. The people who had once guided it were left to find their own way down.

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A. Appendix: Additional Figures and Tables

A.1 Mean Age Over Time

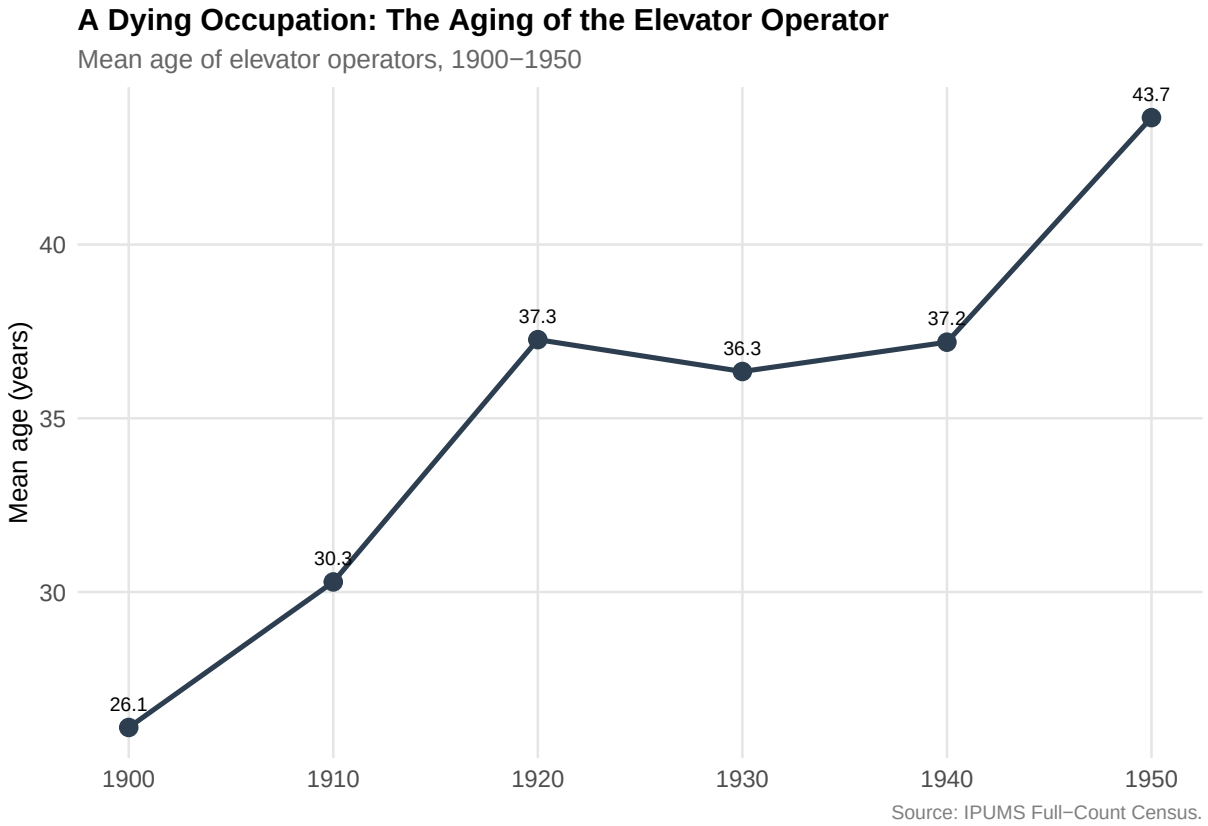

Figure 11: Mean Age of Elevator Operators, 1900–1950

Notes: Mean age of elevator operators at each census date, compared to mean age of all employed workers. Source: IPUMS Full-Count Census.

Table 9: Occupational Transitions by Age Group (1940 Age)

1950 Occupation	20-29	30-39	40-49	50-59	60+	<20
Craftsman	15.5	13.2	10.7	5.7	2.0	17.3
Elevator operator	5.0	16.3	30.3	33.5	17.4	0.7
Farm worker	11.0	12.1	11.5	10.3	6.4	11.1
Not in labor force	12.6	9.8	10.7	23.8	62.5	13.9
Operative/manufacturing	18.2	13.1	8.3	5.0	2.0	19.1
Other	18.2	14.7	9.5	7.2	2.4	16.3

Note:
Column percentages within each age group. Top 6 destination categories shown; remaining categories (clerical/sales, other service, other building service, professional/technical, manager/proprietor, laborer) omitted. Columns do not sum to 100%. Source: IPUMS + MLP v2.0.

A.2 Transition by Age Group

A.3 State-Level Detail

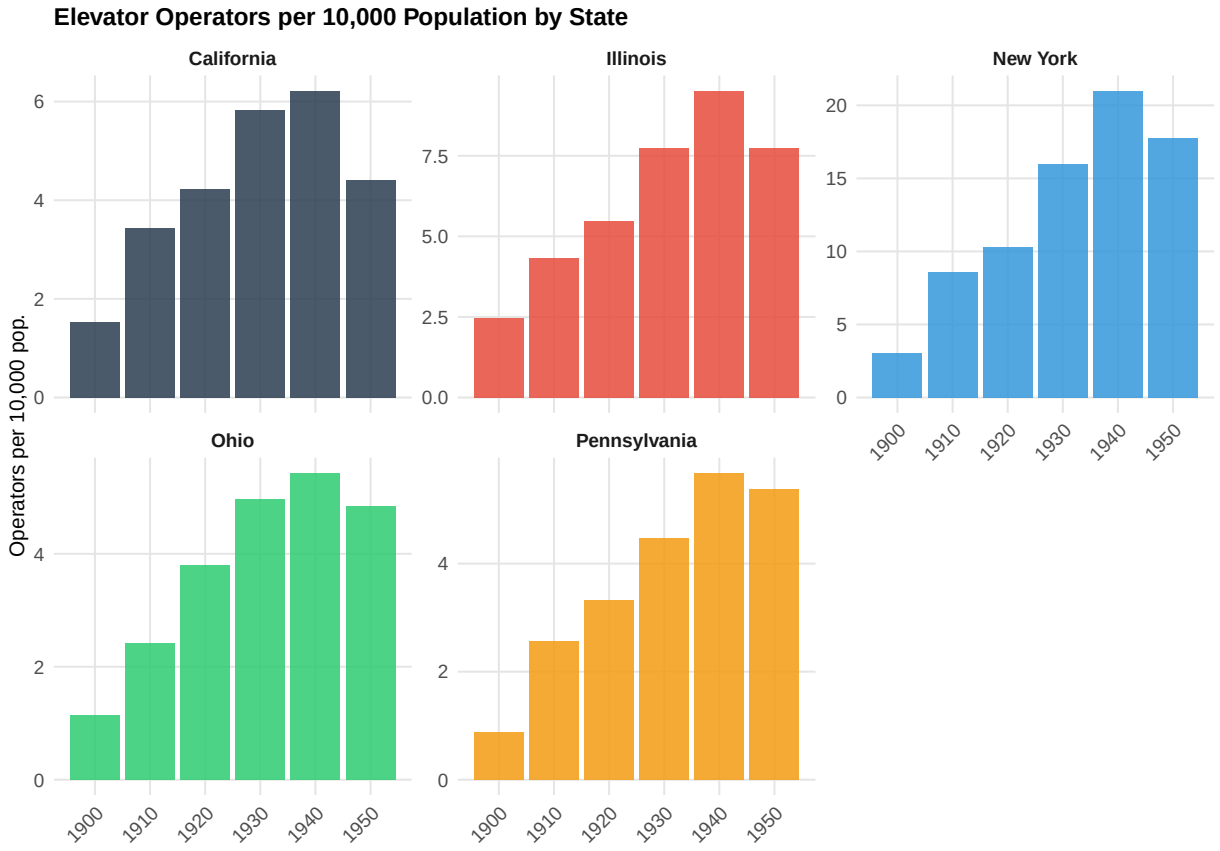

Figure 12: Elevator Operators by State, 1900–1950

Notes: Small multiples showing elevator operators per 10,000 employed in each state across census years. Source: IPUMS Full-Count Census.

B. State-Level Evidence from Synthetic Control

The individual-level analysis in Section 5 establishes that New York City operators were more likely to persist in the occupation than operators elsewhere. This appendix presents corroborating state-level evidence using the synthetic control method (Abadie et al., 2010, 2015), comparing New York State's elevator operator concentration to a weighted combination of donor states.

We construct a synthetic New York State by weighting 18 donor states (those with complete census data across 1900–1950) to match New York's pre-1945 trajectory in elevator operators per 1,000 building service workers. Figure 13 shows that actual New York diverges

above synthetic New York after 1945: the state retained a higher concentration of operators than the synthetic comparison would predict. The permutation p-value is 0.056 (18 placebos), and the post/pre RMSPE ratio ranks New York second among all units.

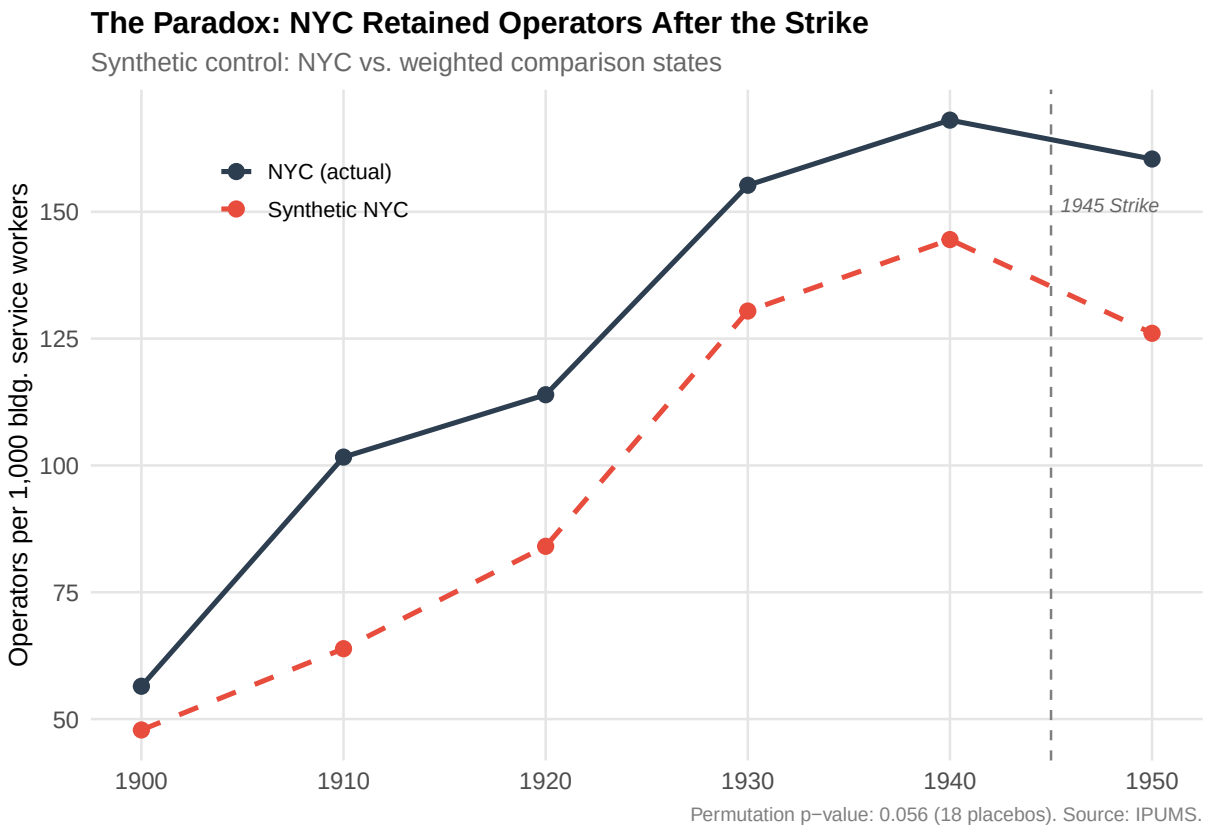

Figure 13: New York vs. Synthetic New York: Elevator Operator Concentration, 1900–1950

Notes: Synthetic control method. Outcome: elevator operators per 1,000 building service workers. Comparison unit: New York State. Donor pool: 18 states with complete 1900–1950 census data. Permutation p-value = 0.056 (18 placebos). Source: IPUMS Full-Count Census.

This state-level pattern is consistent with the individual-level finding that NYC's thick institutional environment—union contracts, building codes, and legacy building stock—slowed displacement relative to other locations. However, several limitations prevent causal interpretation. First, only one post-1945 census observation (1950) is available, precluding assessment of dynamics. Second, the event study (Figure 15) reveals differential pre-trends, with New York's operator concentration converging toward other states throughout 1900–1940. Third, New York State imperfectly proxies the NYC-specific institutional environment. We present this evidence as a structured descriptive comparison that corroborates the individual-level findings, not as a causal estimate of the strike's effect.

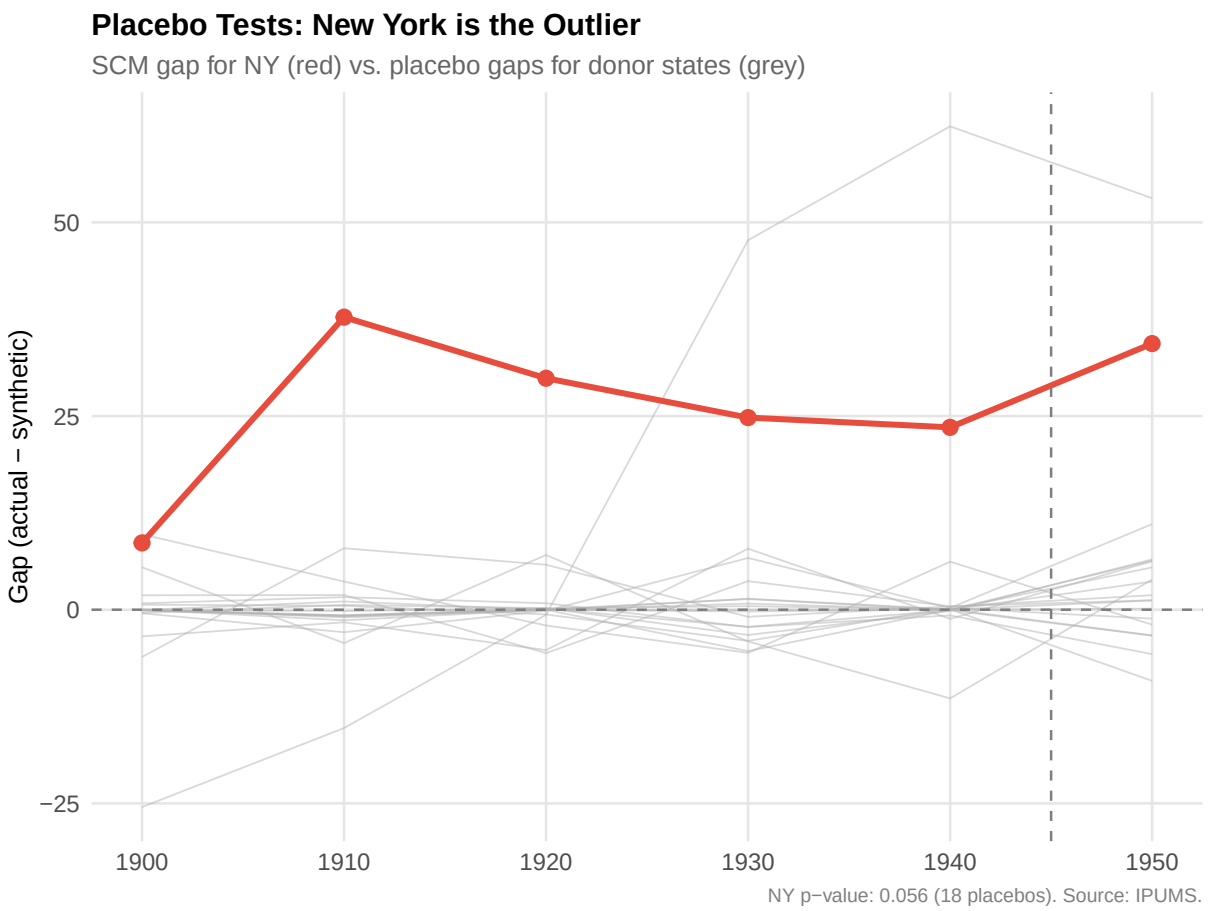

Figure 14: Placebo Tests: Permutation Inference for Synthetic Control

Notes: Each gray line represents a placebo synthetic control, treating a donor state as if it had experienced the 1945 shock. The black line is the actual New York State gap. 18 donor states with complete 1900–1950 data. Source: IPUMS Full-Count Census.

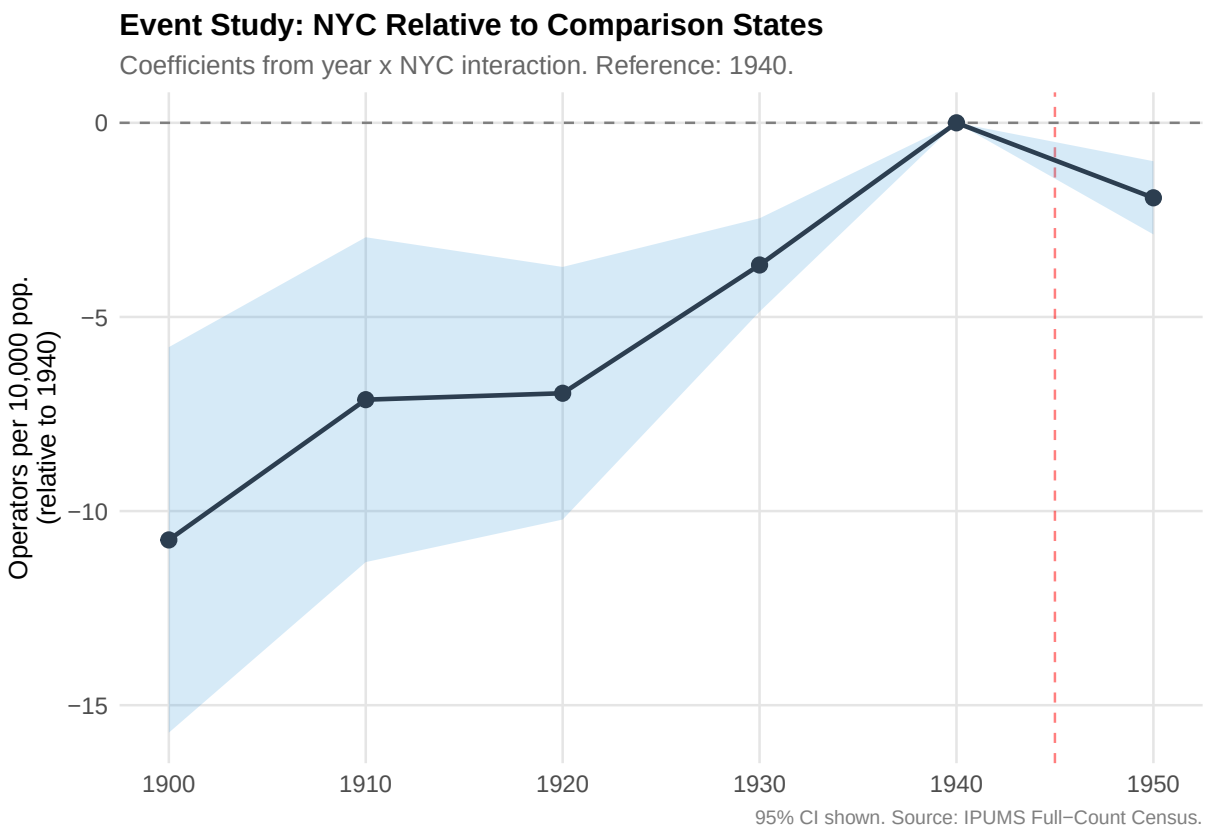

Figure 15: Event Study: New York State Elevator Operators Relative to Other States

Notes: Coefficients from elev_per_10k_pop $= \sum_t \beta_t \cdot \mathbb{1}[\text{NY State}] \times \mathbb{1}[\text{year} = t] + \alpha_s + \gamma_t + \varepsilon_{st}$. Reference year: 1940. Outcome: elevator operators per 10,000 total employed. 95% confidence intervals shown. Source: IPUMS Full-Count Census.

The negative coefficients indicate that New York had fewer elevator operators per 10,000 employed than comparison states in each year, relative to the 1940 baseline. The coefficients' progression from −10.7 (1900) toward −1.9 (1950) shows the gap narrowing steadily over time—consistent with New York's growing concentration of operators outpacing comparison states. The 1950 coefficient (−1.931) represents the smallest gap in the series, indicating that New York's operator workforce contracted *less* than comparison states after 1945, corroborating the individual-level persistence finding.

B.1 SCM Gap Table

B.2 Event Study, Triple Difference, and Comparison Group Robustness

A triple-difference specification—elevator operators versus janitors, New York versus comparison states, before versus after 1945—provides additional evidence by differencing out occupation-specific and state-specific trends simultaneously. The positive and significant

Table 10: New York State vs. Synthetic New York: Elevator Operators per 1,000 Building Service Workers

Year	NY State	Synthetic NY	Gap	Period
1900	56.5	47.9	8.6	Pre-1945
1910	101.6	63.9	37.8	Pre-1945
1920	113.9	84.1	29.9	Pre-1945
1930	155.2	130.4	24.8	Pre-1945
1940	168.1	144.5	23.5	Pre-1945
1950	160.4	126.0	34.4	Post-1945

Note:
Permutation p-value: 0.056 (18 placebos).

coefficient (+0.825, $p < 0.01$) is consistent with the pattern that New York retained operators longer, though the saturated fixed effects in the small panel (120 observations, $R^2 = 0.995$) warrant cautious interpretation.

Acknowledgements

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Project Repository: https://github.com/SocialCatalystLab/ape-papers

Table 11: Robustness Checks

	Event Study	Triple Diff	Excl. Janitors
year = 1900 × NY	-10.743***		
	(2.533)		
year = 1910 × NY	-7.131***		
	(2.134)		
year = 1920 × NY	-6.965***		
	(1.660)		
year = 1930 × NY	-3.662***		
	(0.614)		
year = 1950 × NY	-1.931***		
	(0.481)		
ddd_treat		0.825***	
		(0.071)	
is_elevator_1940			0.035***
			(0.009)
Num.Obs.	60	120	290561
R2	0.856	0.995	0.045

* p <0.1, ** p <0.05, *** p <0.01
Triple diff: elevator vs janitor × NY State × post-1945.